Q10. SQL Query Generation Using Structured Prompting

Aim: To generate an SQL query using a structured prompt and verify its output.

Algorithm:

1. Define the student table structure.
2. Create a structured prompt describing the SQL task.
3. Write an SQL query to select students scoring above 80.
4. Sort the results in descending order of marks.
5. Execute the query and verify the output.

Program:

```
import sqlite3

# Create an in-memory database
conn = sqlite3.connect(":memory:")
cursor = conn.cursor()

# Create Students table
cursor.execute("""
CREATE TABLE Students (
    student_id INTEGER PRIMARY KEY,
    name TEXT,
    department TEXT,
    marks INTEGER
)
""")

# Insert sample records
students = [
    (1, "Arun", "CSE", 90),
    (2, "Priya", "IT", 85),
    (3, "Kumar", "ECE", 70),
    (4, "Divya", "CSE", 95)
]
cursor.executemany("INSERT INTO Students VALUES (?, ?, ?, ?)", students)

# Structured prompt
prompt = """
Task: Select student names and marks.
Condition: Marks must be greater than 80.
Sorting: Marks in descending order.
"""

print("Structured Prompt:")
print(prompt)
```

```python
print("Structured Prompt:")
print(prompt)

# SQL query
query = """
SELECT name, marks
FROM Students
WHERE marks > 80
ORDER BY marks DESC;
"""

print("Generated SQL Query:")
print(query)

# Execute and display results
cursor.execute(query)
results = cursor.fetchall()

print("Query Output:")
for row in results:
    print(row)

# Validate results
assert all(marks > 80 for name, marks in results)
assert results == sorted(results, key=lambda row: row[1], reverse=True)

print("Query validation successful!")

conn.close()
```

Output:

```
Structured Prompt:

Task: Select student names and marks.
Condition: Marks must be greater than 80.
Sorting: Marks in descending order.

Generated SQL Query:

SELECT name, marks
FROM Students
WHERE marks > 80
ORDER BY marks DESC;

Query Output:
('Divya', 95)
('Arun', 90)
('Priya', 85)
Query validation successful!
```

Result: The SQL query was executed successfully and returned students who scored above 80, sorted by marks in descending order.